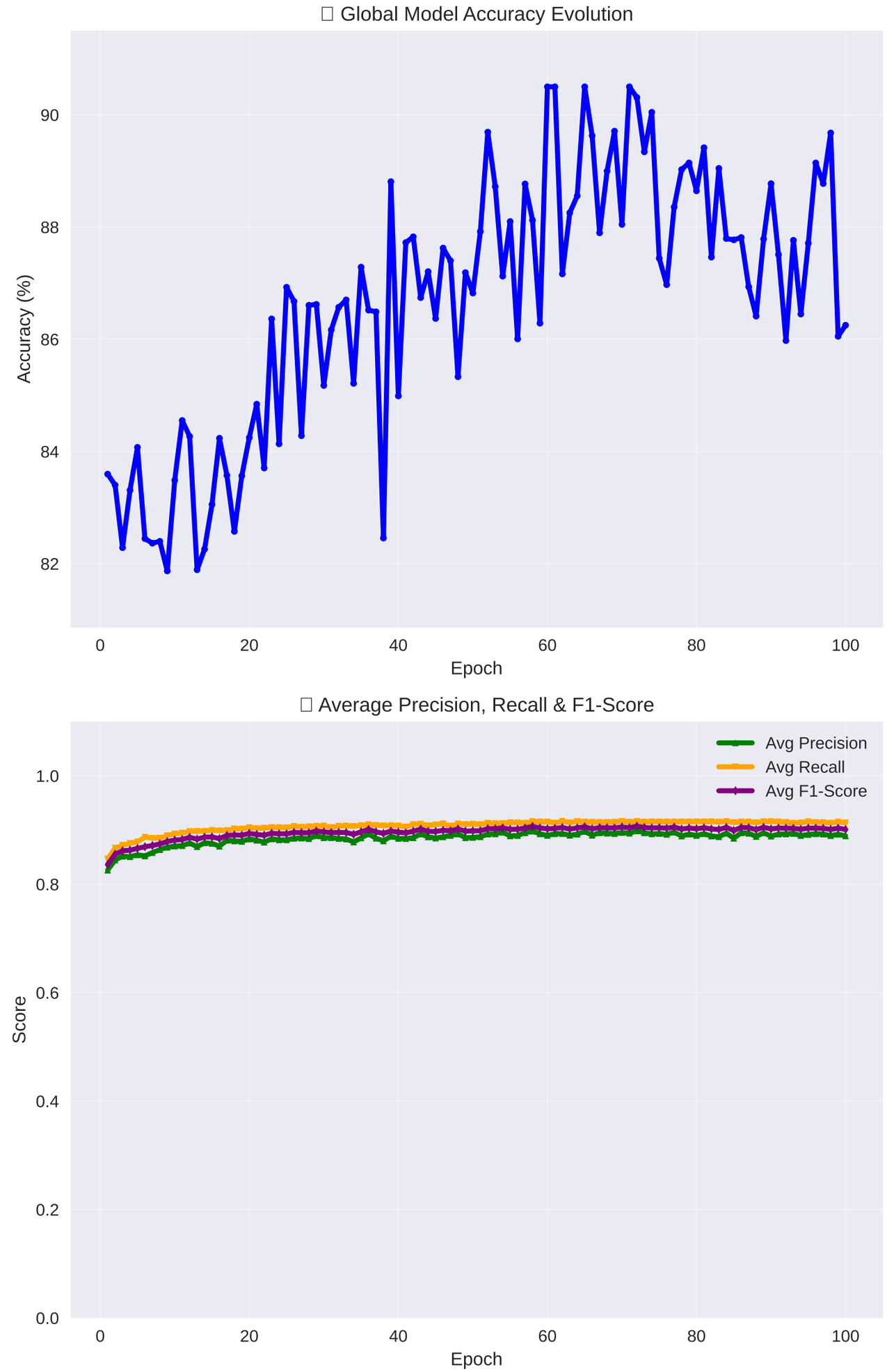
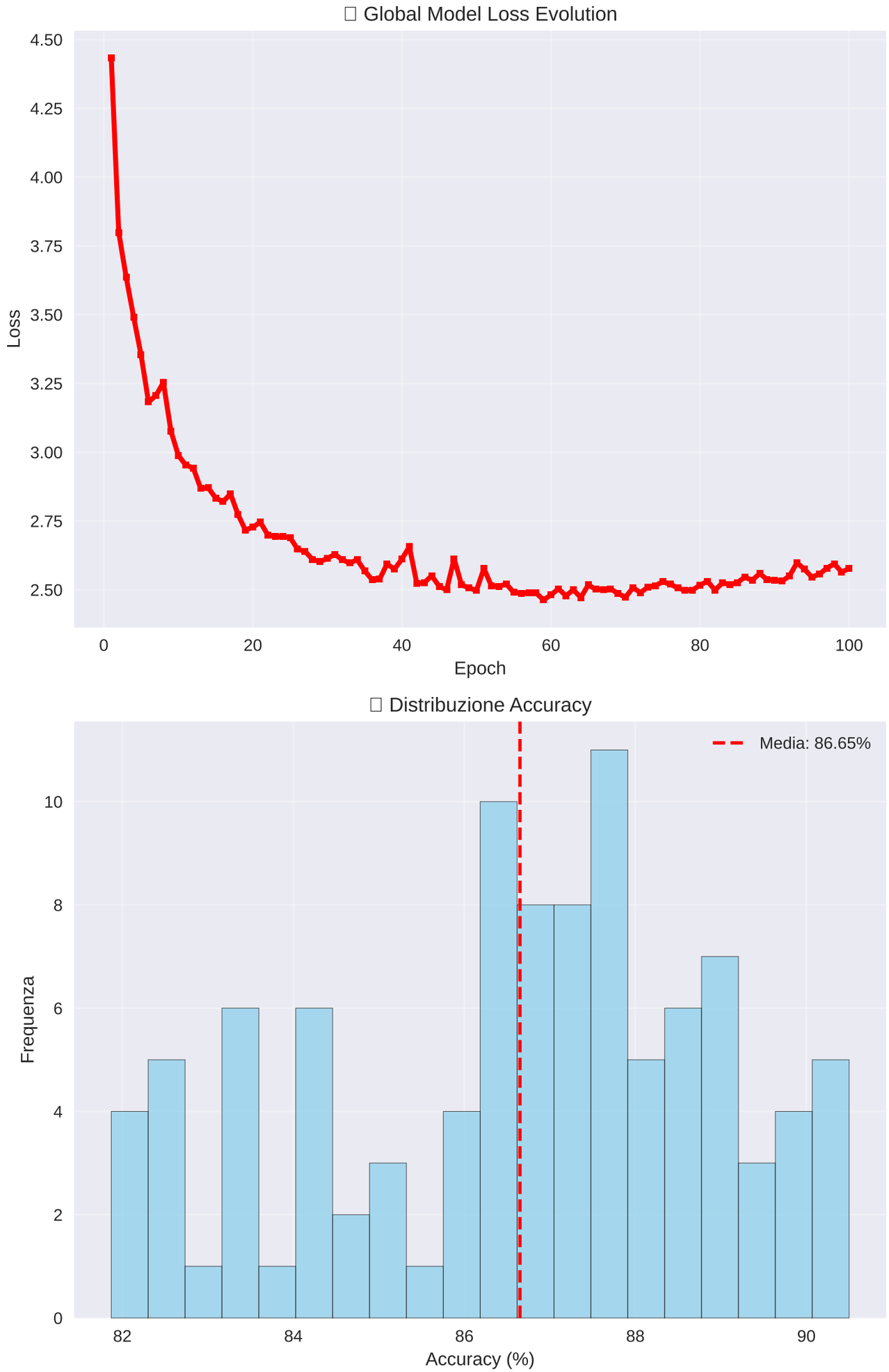
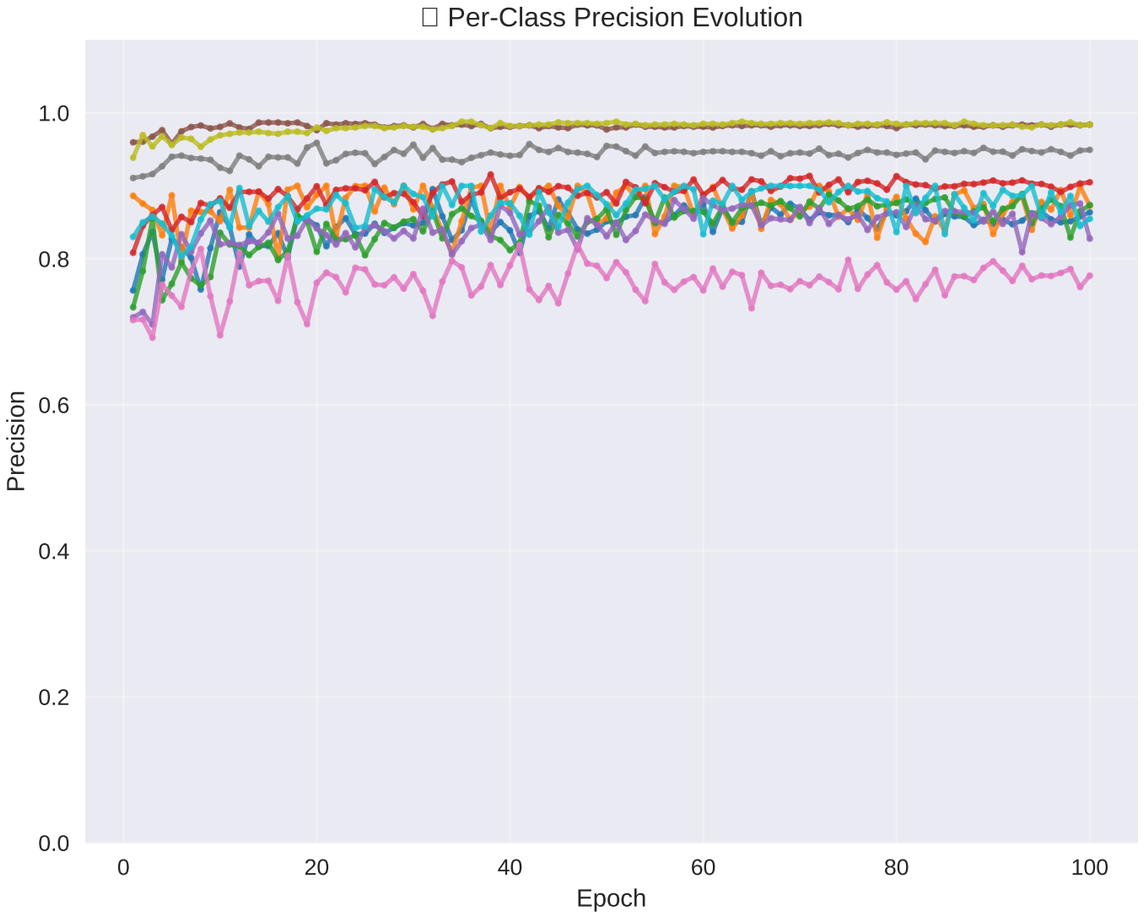
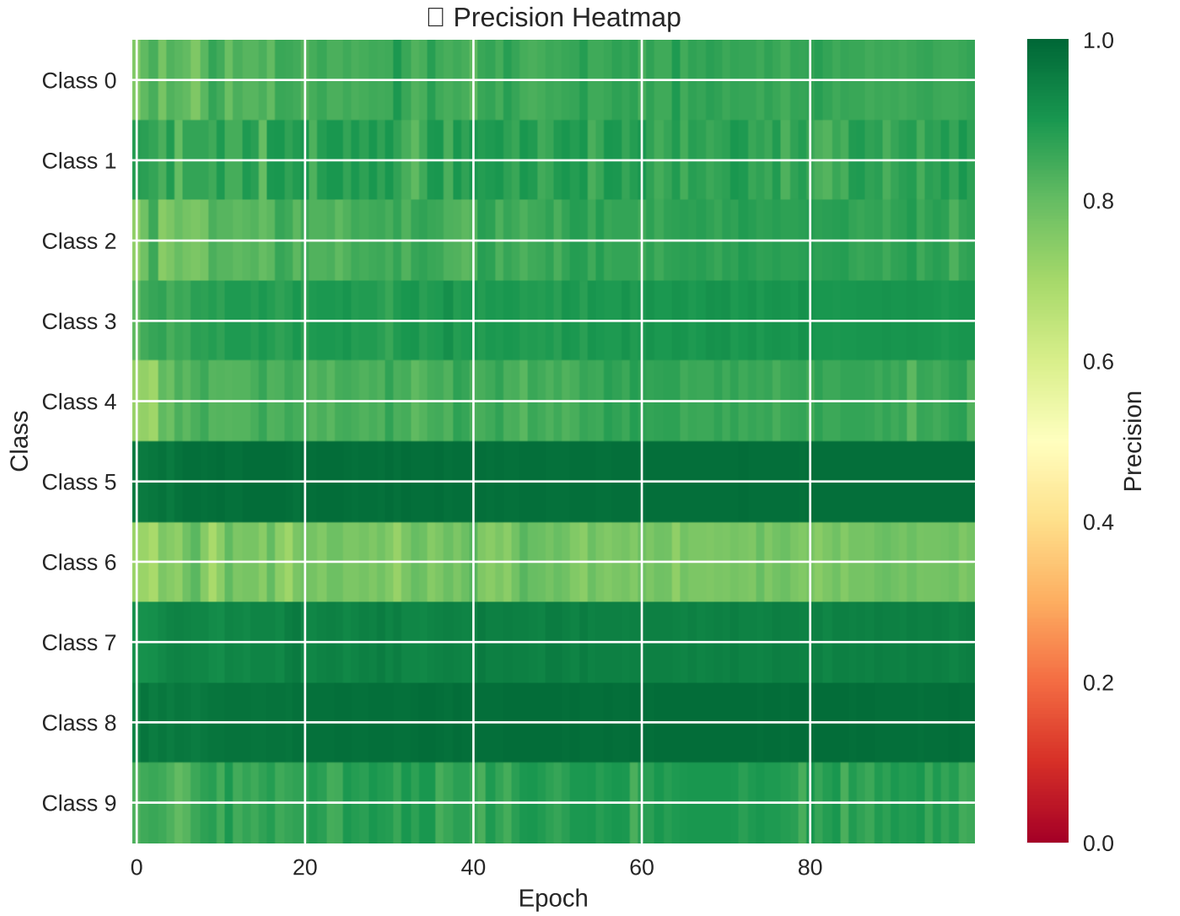
Federated Learning - Main Metrics Overview

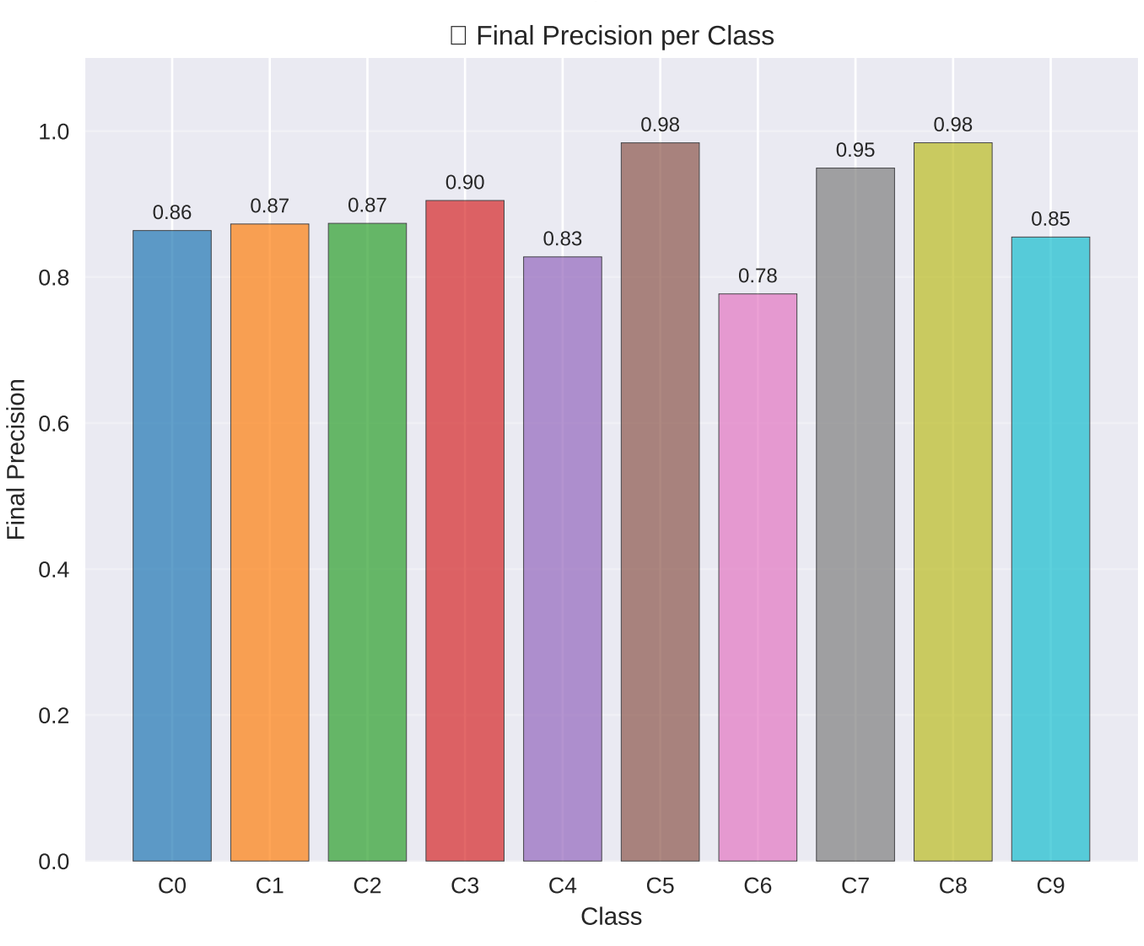


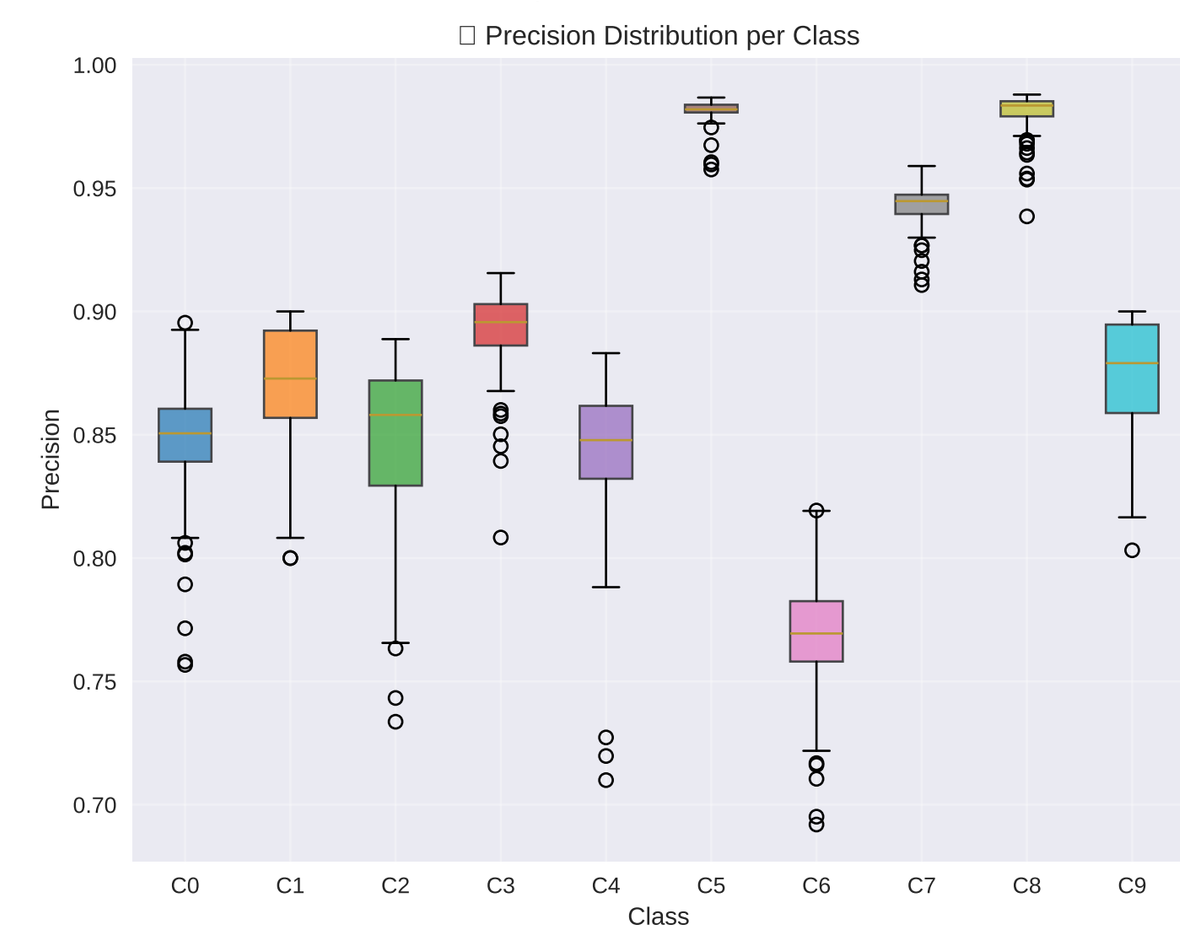


Per-Class Precision Analysis



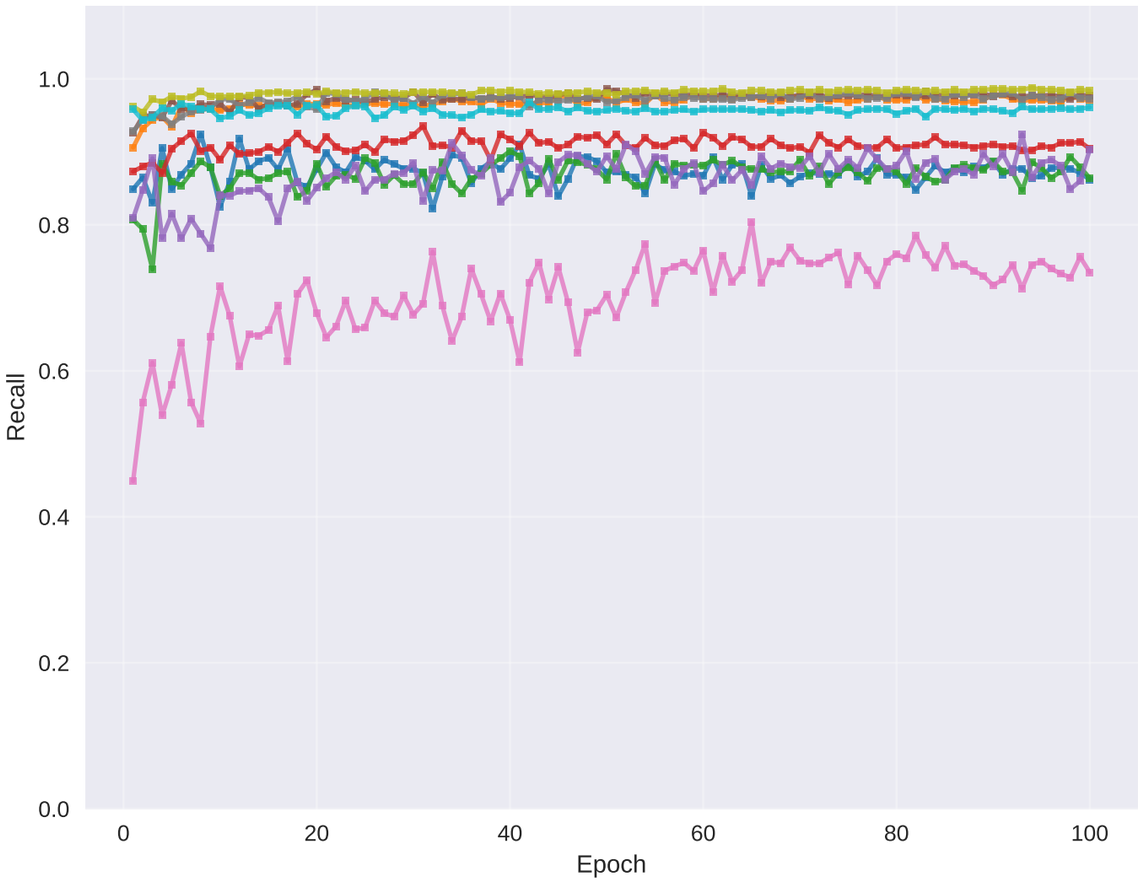




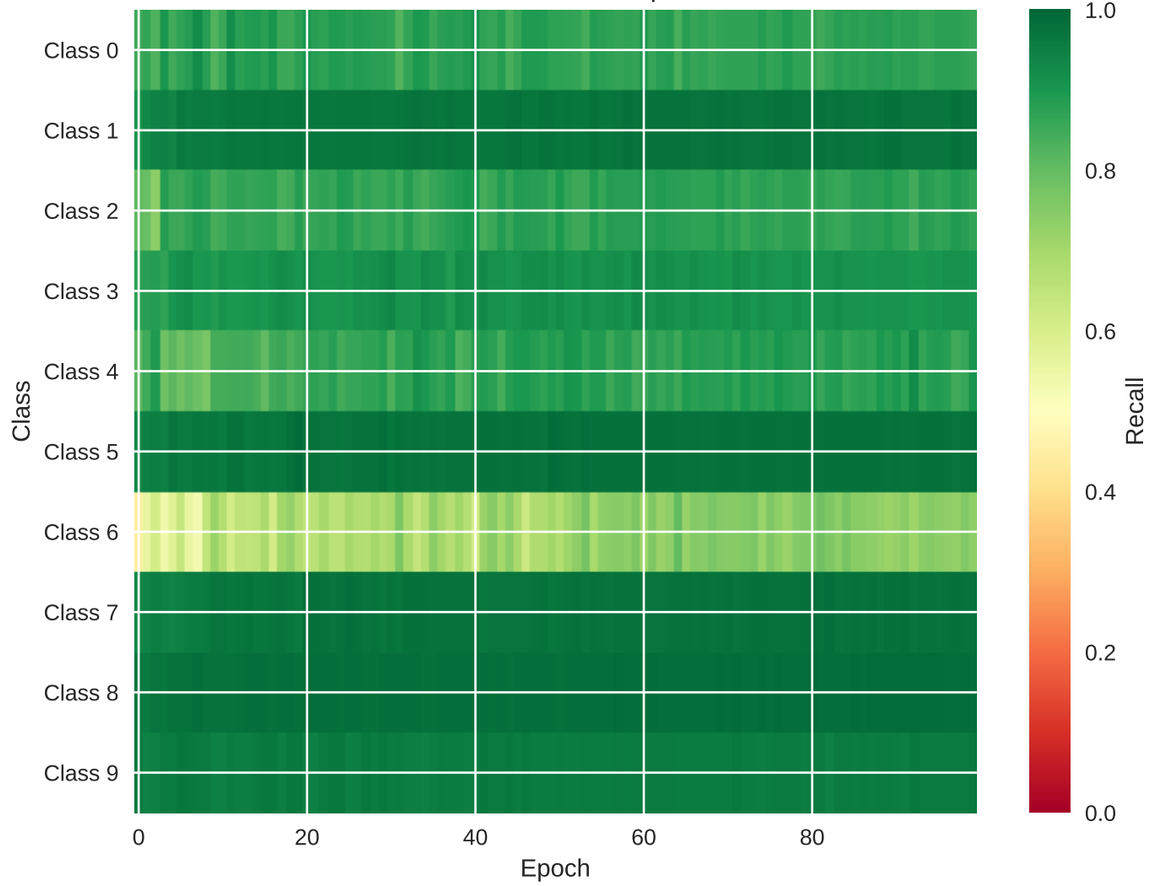


Per-Class Recall Analysis

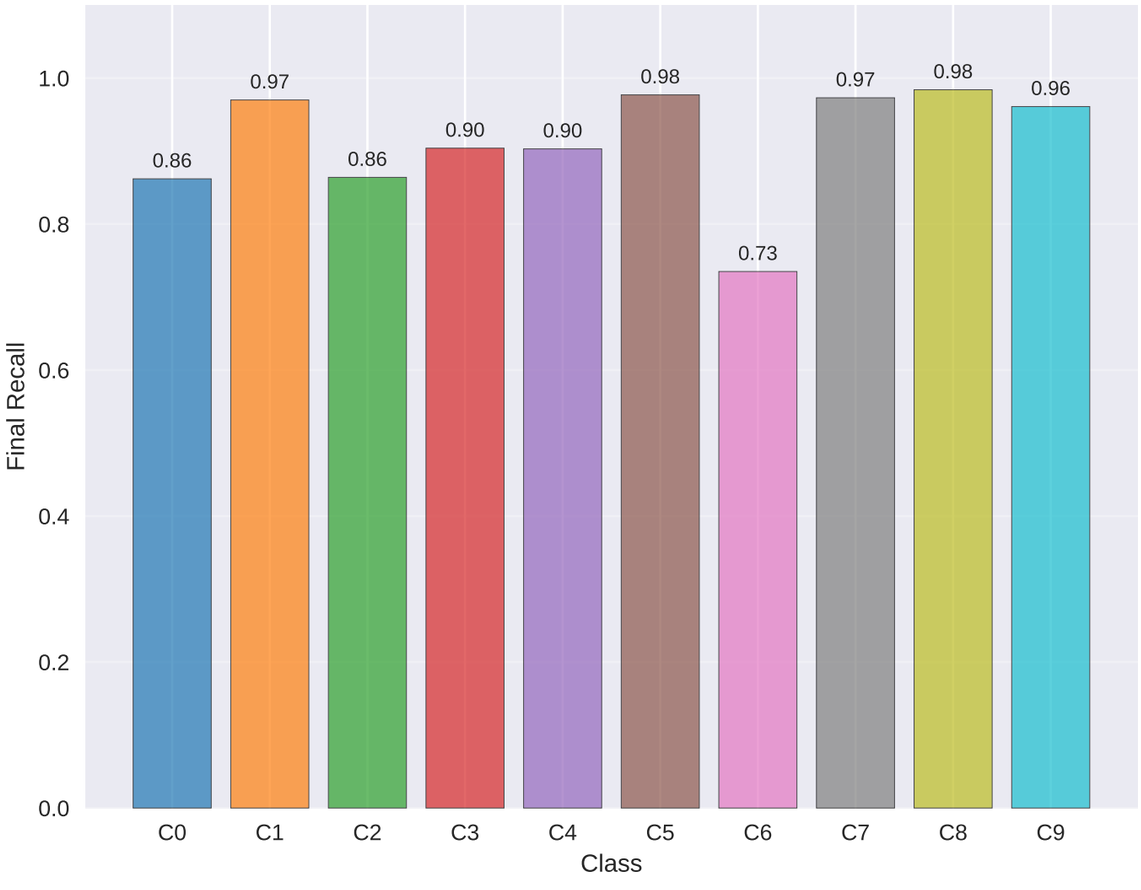
📈 Per-Class Recall Evolution



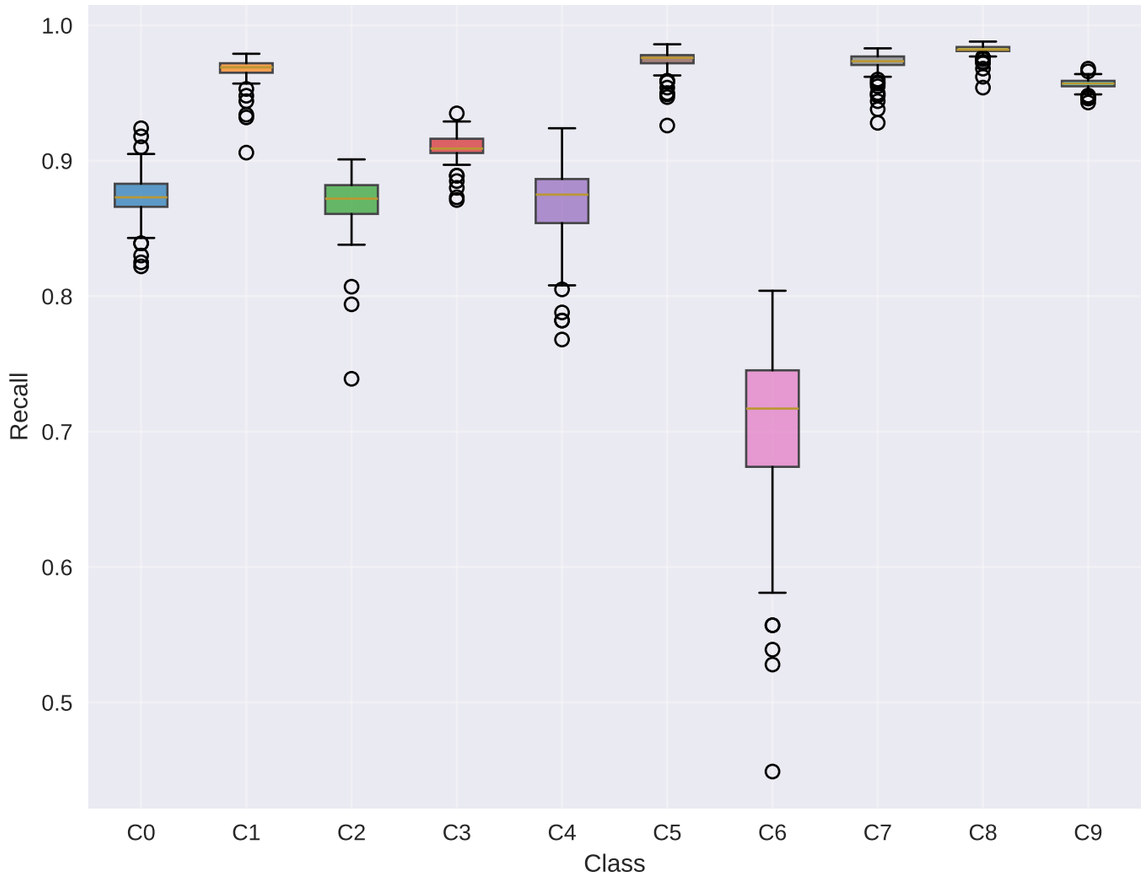
🔥 Recall Heatmap



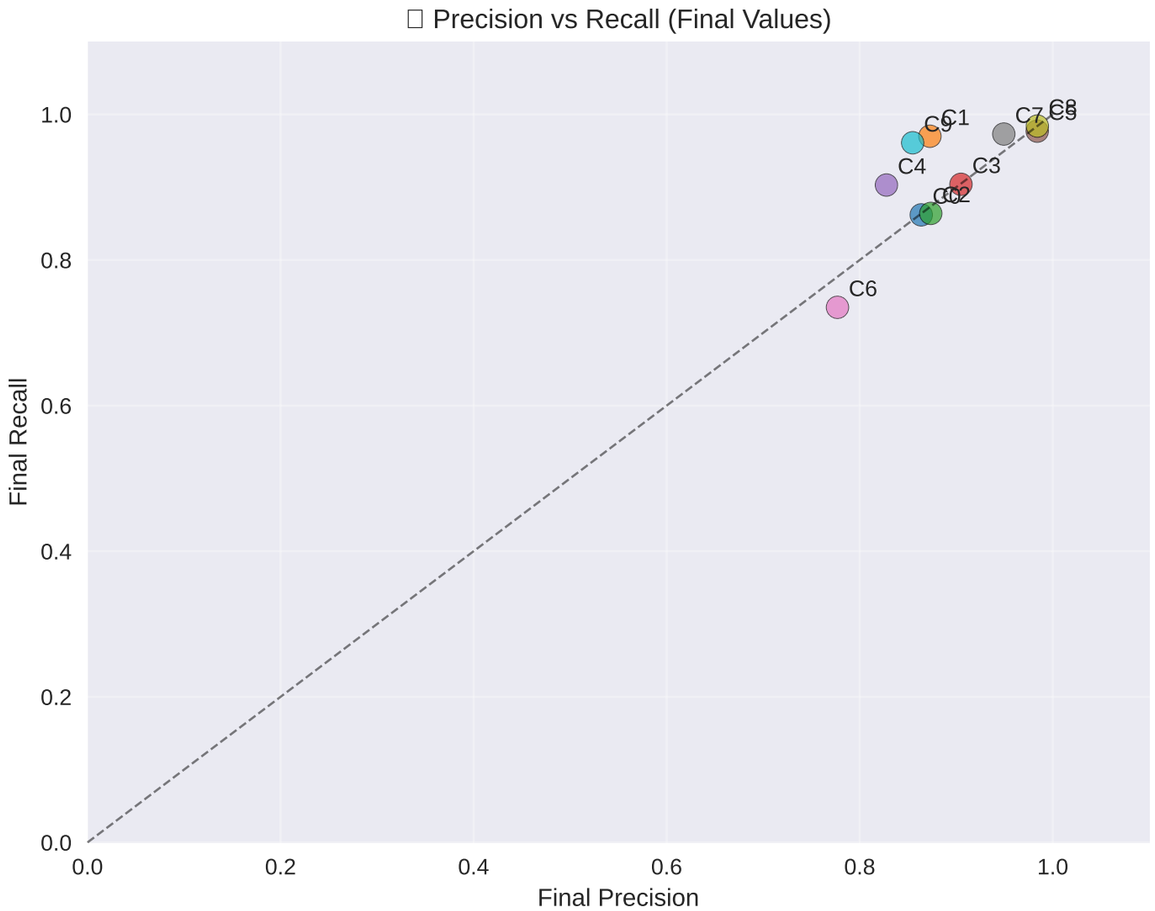
📊 Final Recall per Class



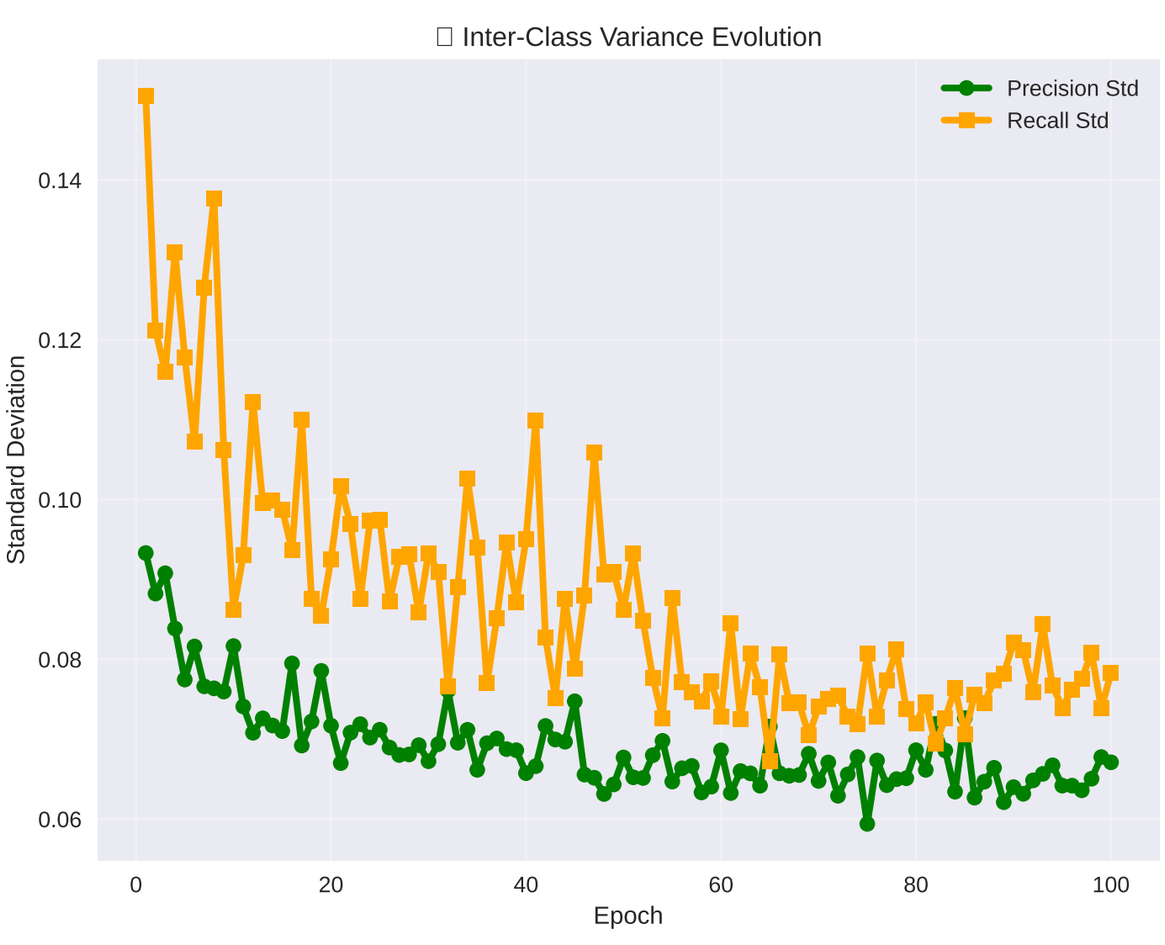
📦 Recall Distribution per Class

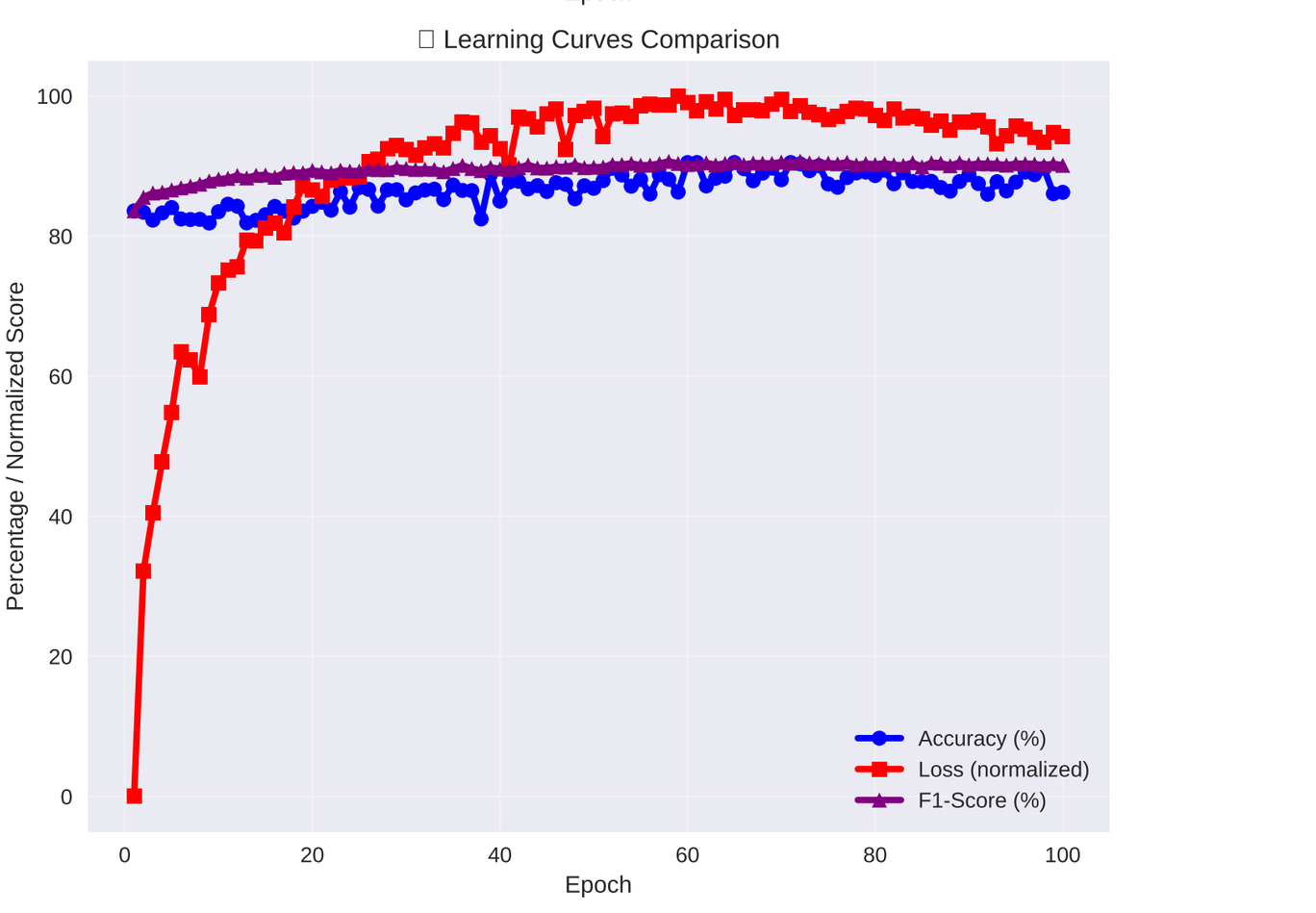


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-05 16:01:37
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 86.25%
• Loss: 2.5768
• Average Precision: 0.889
• Average Recall: 0.913
• Average F1-Score: 0.901

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +2.65%
• Riduzione Loss: +1.8568
• Stabilità Accuracy (CV): 0.0268

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 8 (0.984)
• Peggiore Precision: Classe 6 (0.777)
• Migliore Recall: Classe 8 (0.984)
• Peggiore Recall: Classe 6 (0.735)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.864, R=0.862, F1=0.863
• Classe 1: P=0.873, R=0.970, F1=0.919
• Classe 2: P=0.874, R=0.864, F1=0.869
• Classe 3: P=0.905, R=0.904, F1=0.904
• Classe 4: P=0.828, R=0.903, F1=0.864
• Classe 5: P=0.984, R=0.977, F1=0.980
• Classe 6: P=0.777, R=0.735, F1=0.755
• Classe 7: P=0.949, R=0.973, F1=0.961
• Classe 8: P=0.984, R=0.984, F1=0.984
• Classe 9: P=0.855, R=0.961, F1=0.905

VALUTAZIONE QUALITATIVA:
• Performance Globale: MEDIA
• Stabilità Training: MODERATAMENTE STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
• Considerare aumento epoche o tuning iperparametri
```